

Shiv Kumar

Shivscribe@gmail.com | +91-8685837036 | LinkedIn | GitHub

SUMMARY

Computer Science final-year student at NIT Kurukshetra with hands-on experience building full-stack web applications, REST APIs, and backend systems using the MERN stack, PHP CodeIgniter, Node.js, and MySQL with strong fundamentals in DSA, OOP, DBMS, and Operating Systems.

EDUCATION

National Institute of Technology, Kurukshetra	2022 – 2026
B.Tech in Computer Science	
RED Sr. Sec. School, Jhajjar, Haryana – Intermediate (CBSE)	2021 – 2022
Sarvodya School of Science, Charkhi Dadri, Haryana – Matriculation (CBSE)	2019 – 2020

EXPERIENCE

NHPC Limited – Software Engineering Intern Jun 2025 – Jul 2025

- Worked on a **role-based content management system** in **PHP CodeIgniter** and **MySQL** supporting 4 user roles, with **REST APIs** powering authentication, user management, and admin operations.
- Set up **JWT authentication** along with **role-based access control**, and built out the full content lifecycle submission, admin review, feedback on rejection, resubmission, and final publishing.
- Tuned slower SQL queries and reworked backend logic to bring average API response times down by roughly **35%**, and added a rich text editor with a basic AI-based comment moderation feature.

PROJECTS

Job Adda — MERN Stack Job Portal [GitHub] React, Node.js, Express, MongoDB, JWT

- Built a full-stack job portal with separate role-based dashboards for recruiters and job seekers, covering job posting, search, filtering, and the full application flow, with a responsive React frontend.
- Wrote **REST APIs** in Node.js and Express for jobs, users, applications, and authentication, secured using **JWT** and **role-based access control** for multi-role access.
- Added **indexing and pagination** on MongoDB, making large job listings load close to **40% faster** on bigger result sets.

Multithreaded File Server TCP Sockets, Multithreading

- Built a concurrent TCP file server in C++ that handles clients, mainly to get hands-on with socket programming and core OS concepts.
- Used **mutex-based synchronization** to keep shared resources safe across threads and avoid race conditions during concurrent file operations.
- Tuned thread scheduling and I/O handling, ending up with about **30% better throughput** compared to a single-threaded baseline.

Smart Parking System Python, YOLO, OpenCV

- Built a real-time parking system using **YOLO** and **OpenCV** to detect vehicles and map slot occupancy from a live video feed.
- Added a nearest-slot recommendation using Euclidean distance so drivers are directed to the closest empty spot.
- Optimized the video processing pipeline to run at **20+ FPS** on a regular laptop, making it usable for real-time scenarios.

SKILLS

Languages: Java, Python, C++, C, JavaScript, PHP, SQL

Frontend: HTML, CSS, JavaScript, React.js

Backend: Node.js, Express.js, PHP (CodeIgniter), REST APIs, JWT, RBAC

Databases: MySQL, PostgreSQL, MongoDB

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Tools & Platforms: Git, GitHub, Linux, Postman, VS Code, Jupyter

Other: Basic AWS, CI/CD fundamentals, Docker (learning)

POSITIONS OF RESPONSIBILITY & ACHIEVEMENTS

- **Secretary, Startup Cell, NIT Kurukshetra**: helped organize Smart India Hackathon at the campus level with **400+** participants.
- **Organizing & Video Editing Team, TEDxNITKKR**: coordinated event operations and post-event video production.
- **Creative Lead, Photog (Photography Club), NIT Kurukshetra**: handled team coordination, event coverage, and mentored juniors.
- **1st Place in Cinematography, Confluence NITKKR**.